DATA SCIENCE PROJECT

Predicting MBTI Personality Types

Utilizing Cluster-Based Embeddings and Multi-Level Attention Mechanisms

PRESENTER
Lead Data Scientist

TOPIC
Neural Architecture & MBTI

PROBLEM STATEMENT & CONTEXT

MBTI Dimensions

Classifying users across 4 binary axes: **E/I** (Introversion), **N/S** (Intuition), **T/F** (Thinking), and **J/P** (Judging).

The Data Challenge

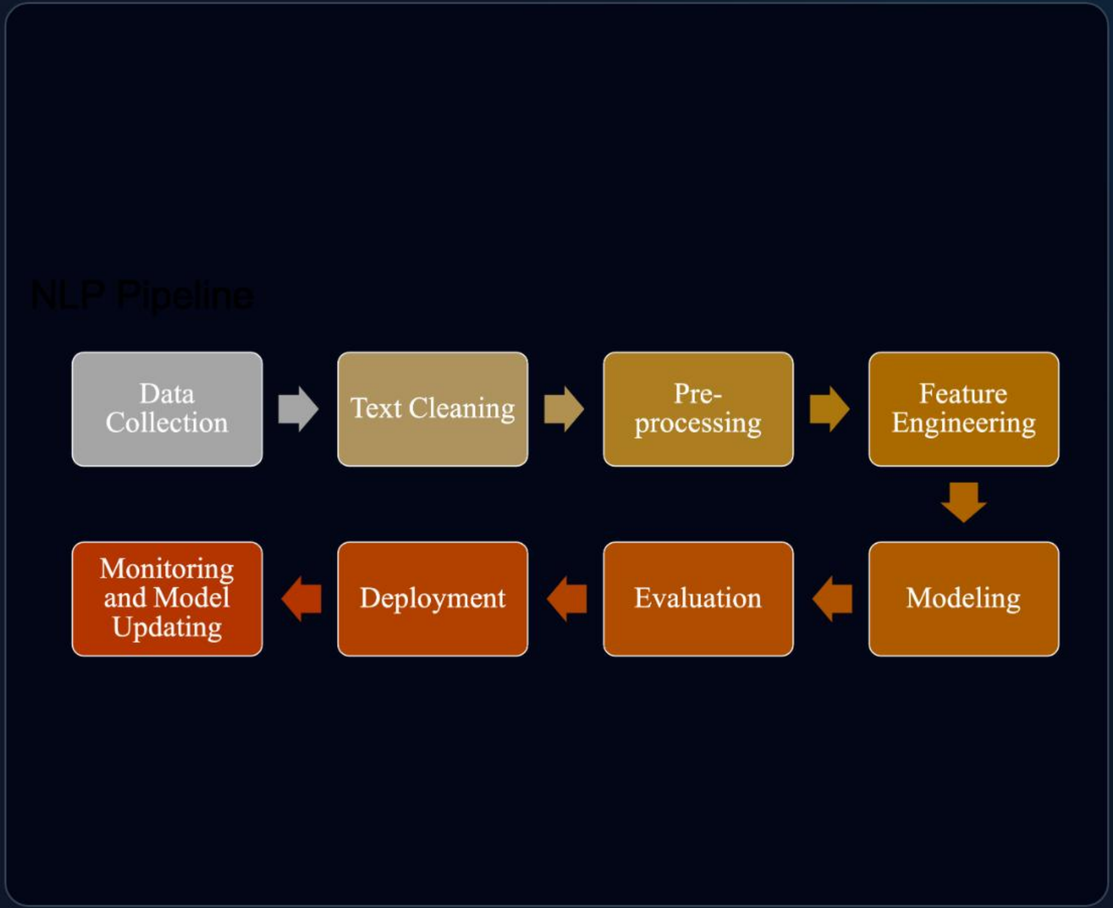
Social text is noisy. Each user provides ~50 posts. The goal is to extract deep psychological markers from fragmented, informal content.

DATA PIPELINE & PREPROCESSING

Intensive Text Cleaning

Transforming raw social media noise into structured tokens using **SpaCy** and **NLTK**.

- ✓ Removal of URLs, numbers, and special chars
- ✓ Contraction mapping & normalization
- ✓ Advanced Lemmatization via SpaCy



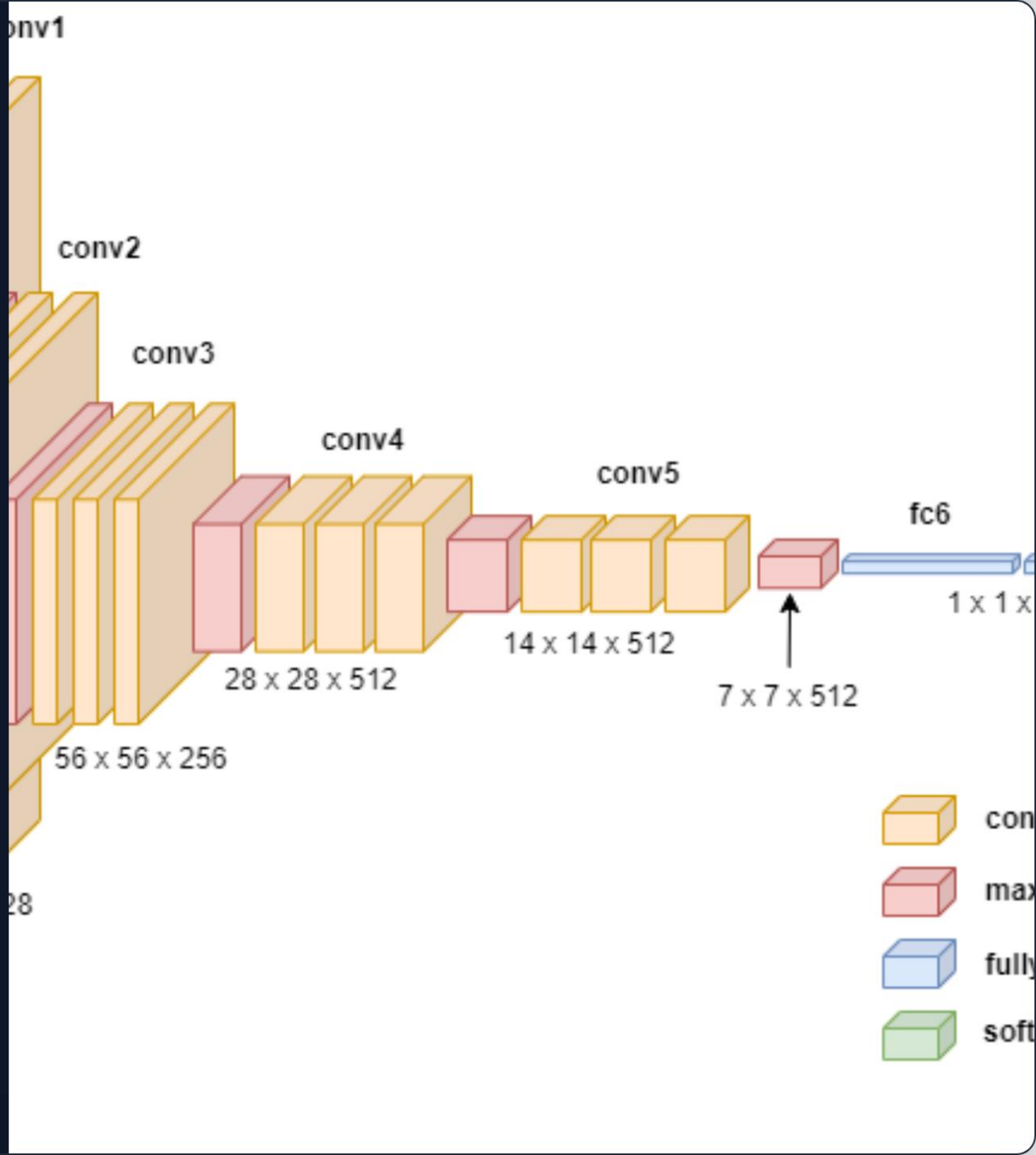
LEVEL 1 & 2: POST AGGREGATION

Level 1: Word-Level Bi-LSTM

Uses 100-D GloVe embeddings. The Bi-LSTM processes sequential dependencies in each post to output 50 post vectors per user.

Level 2: Attention Aggregator

Instead of a mean, we learn dynamic weights for each post. The model focuses on posts that contain the highest personality "signal."



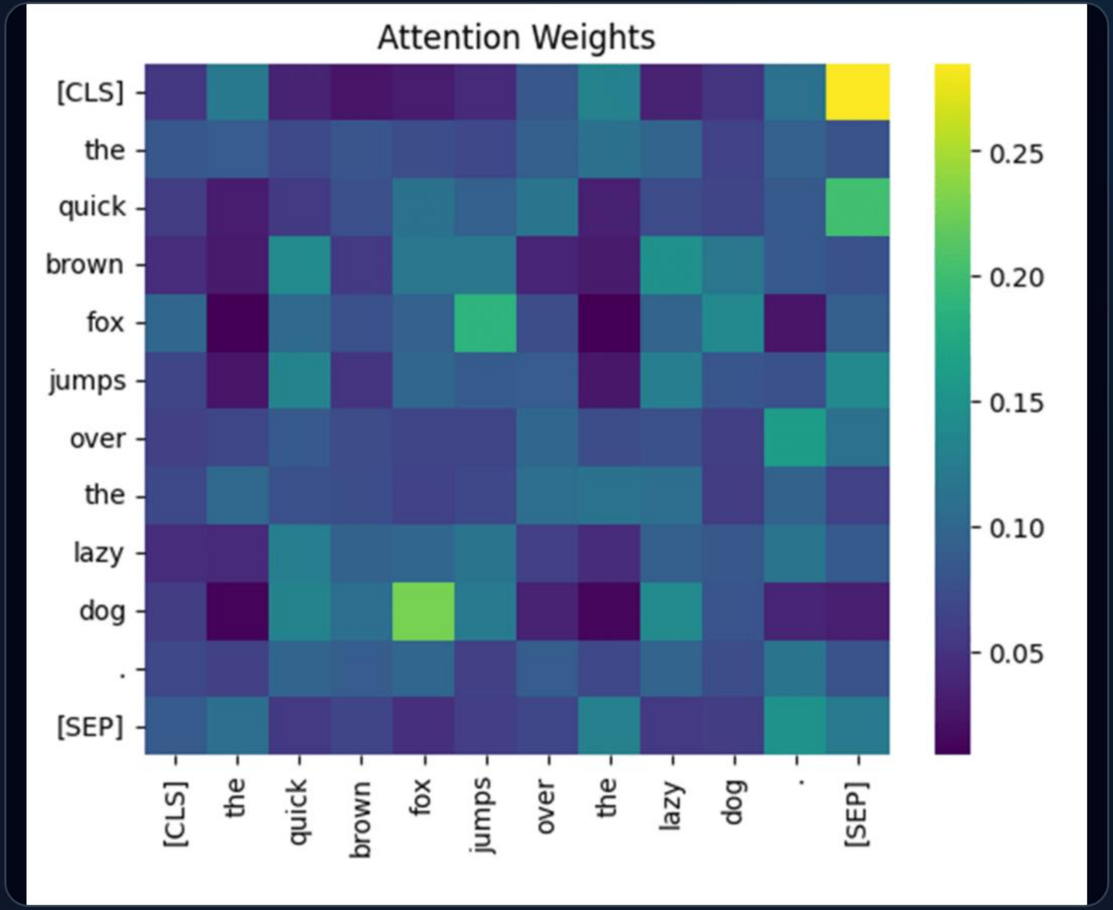
LEVEL 2.5: CLUSTER ATTENTION

The Core Innovation

We initialize 16 learnable **Prototype Vectors** representing the 16 MBTI types.

The Person Vector acts as a **Query**, attending to these prototypes to create a **Cluster Context Vector**.

This captures the user's soft alignment with existing psychological archetypes.



FULL MODEL ARCHITECTURE

Level 1: Maps words to GloVe vectors and creates post summaries.

Level 2.5: Attends to 16 personality prototypes (The Innovation).

Level 3: Concatenates Person + Cluster vectors for binary classification.

WHY CLUSTER ATTENTION?

Archetypal Contrast
By comparing users to prototypes, the model identifies "distinguishing" traits more clearly than flat embeddings.

Contextual Depth
The Cluster Context Vector acts as a 'global anchor,' providing a psychological frame of reference for the classifier.

Feature Fusion
Final classification relies on both the individual signal (Person Vector) and the group signal (Cluster Vector).

TRAINING & CLASS IMBALANCE

BCE

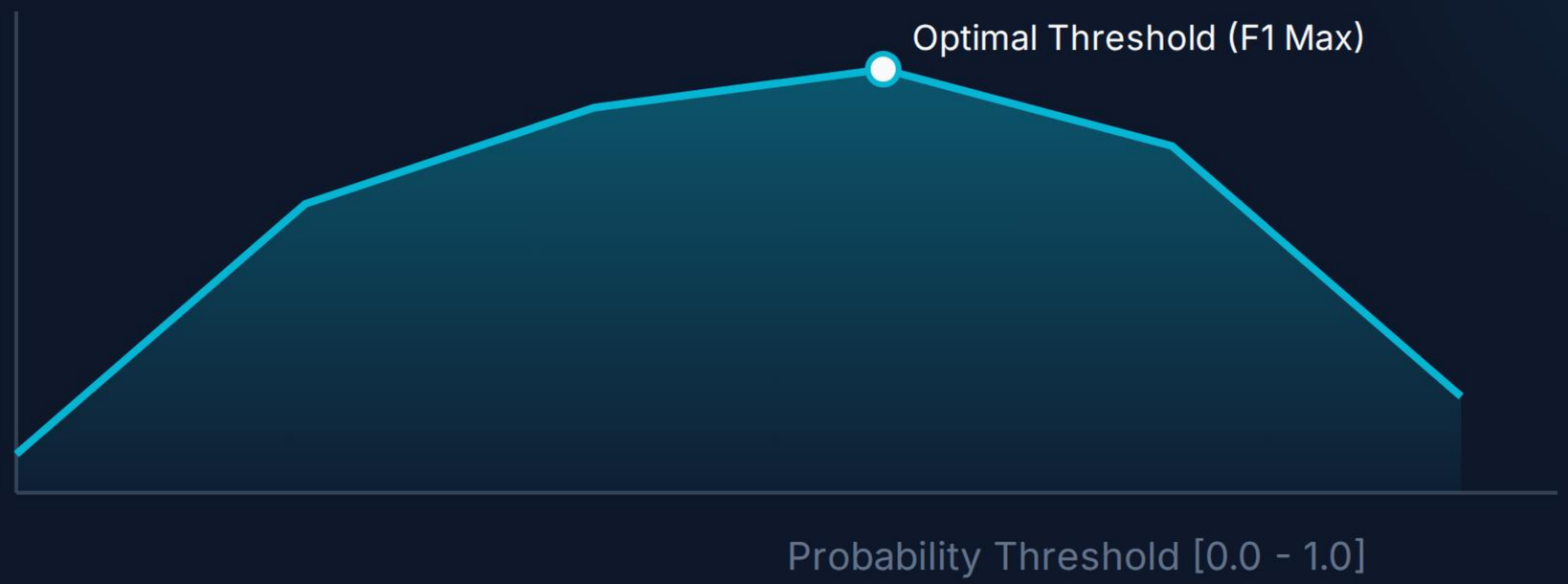
With Logits Loss

Imbalance Mitigation

MBTI data is notoriously skewed (e.g., Introverts dominate online forums).

- Computed **Positive Weights** applied dynamically.
- Adam Optimizer with **ReduceLROnPlateau**.
- Dropout & ReLU in the final head for stability.

THRESHOLD OPTIMIZATION



Rather than a static 0.5 cutoff, we calibrate thresholds per dimension to maximize F1-Scores on the evaluation set.

Conclusion

A New Standard for Psychological Text Analysis

Achieved Metrics

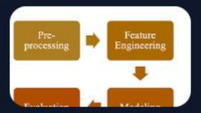
High F1-scores across 4 dimensions and robust 4/4 exact type accuracy through soft-prototype alignment.

Future Work

Exploring Transformer-based (BERT/RoBERTa) word embeddings to replace GloVe for richer contextual cues.

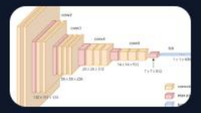
Questions? Let's dive deeper into the Level 2.5 Logic.

IMAGE SOURCES



https://alvinntnu.github.io/NTNU_ENC2045_LECTURES/_images/nlp-pipeline.png

Source: alvinntnu.github.io



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